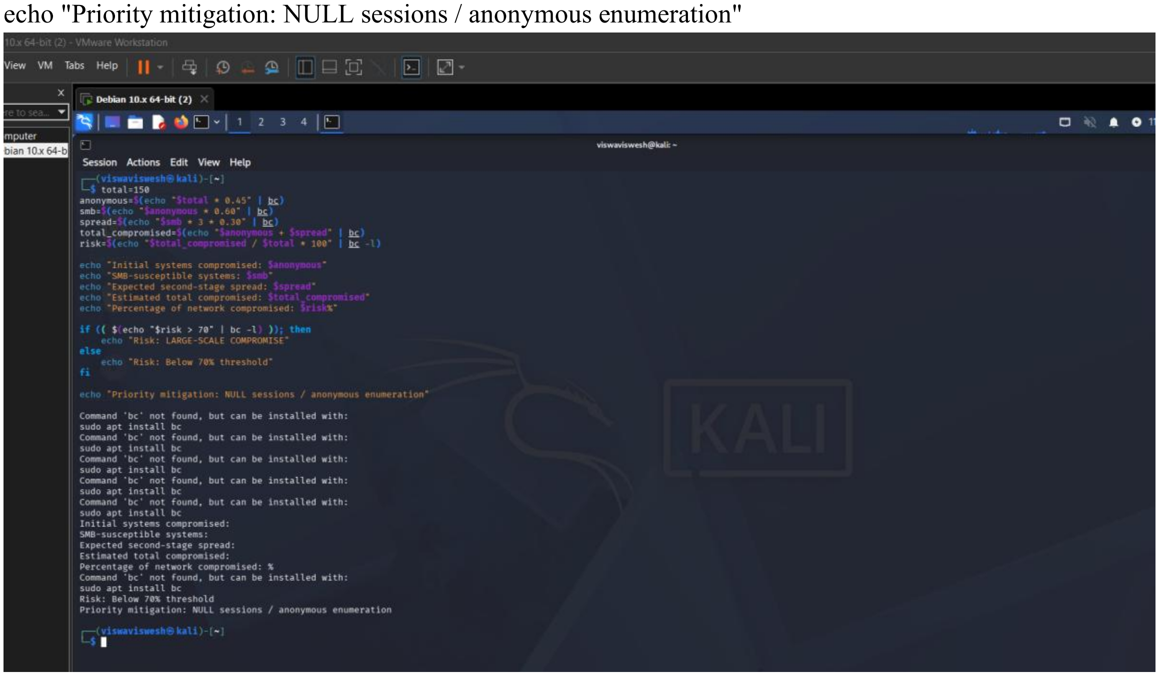
Q1. #!/bin/bash
total=150
anonymous=$(echo "$total * 0.45" | bc)
smb=$(echo "$anonymous * 0.60" | bc)
spread=$(echo "$smb * 3 * 0.30" | bc)
total_compromised=$(echo "$anonymous + $spread" | bc)
risk=$(echo "$total_compromised / $total * 100" | bc -l)

echo "Initial systems compromised: $anonymous"
echo "SMB-susceptible systems: $smb"
echo "Expected second-stage spread: $spread"
echo "Estimated total compromised: $total_compromised"
echo "Percentage of network compromised: $risk%"

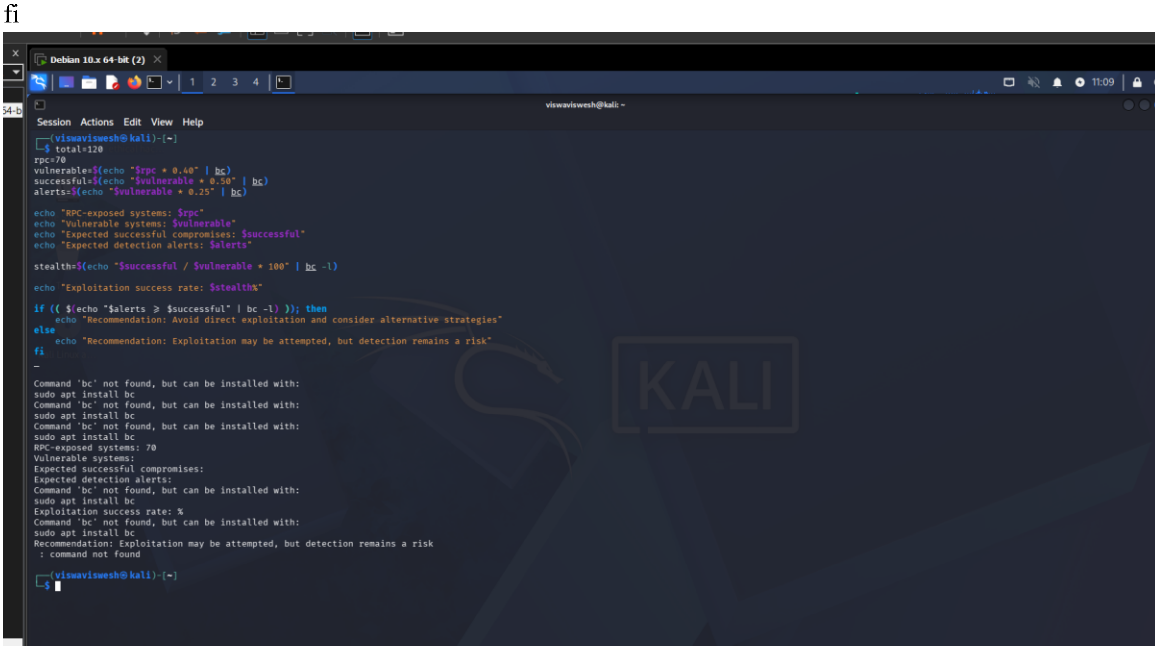
if (($(echo "$risk > 70" | bc -l))); then
    echo "Risk: LARGE-SCALE COMPROMISE"
else
    echo "Risk: Below 70% threshold"
fi

echo "Priority mitigation: NULL sessions / anonymous enumeration"



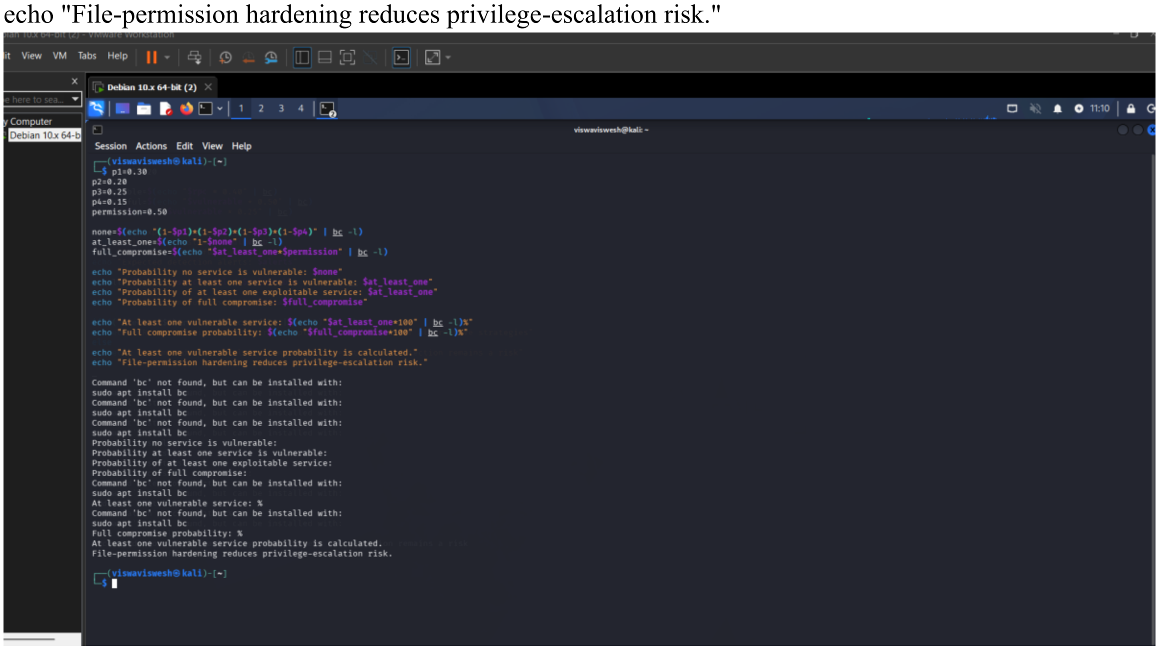
Q2. #!/bin/bash

```
total=120
rpc=70
vulnerable=$(echo "$rpc * 0.40" | bc)
successful=$(echo "$vulnerable * 0.50" | bc)
alerts=$(echo "$vulnerable * 0.25" | bc)

echo "RPC-exposed systems: $rpc"
echo "Vulnerable systems: $vulnerable"
echo "Expected successful compromises: $successful"
echo "Expected detection alerts: $alerts"

stealth=$(echo "$successful / $vulnerable * 100" | bc -l)

echo "Exploitation success rate: $stealth%"

if (( $(echo "$alerts >= $successful" | bc -l) )); then
   echo "Recommendation: Avoid direct exploitation and consider alternative strategies"
else
   echo "Recommendation: Exploitation may be attempted, but detection remains a risk"
fi
```



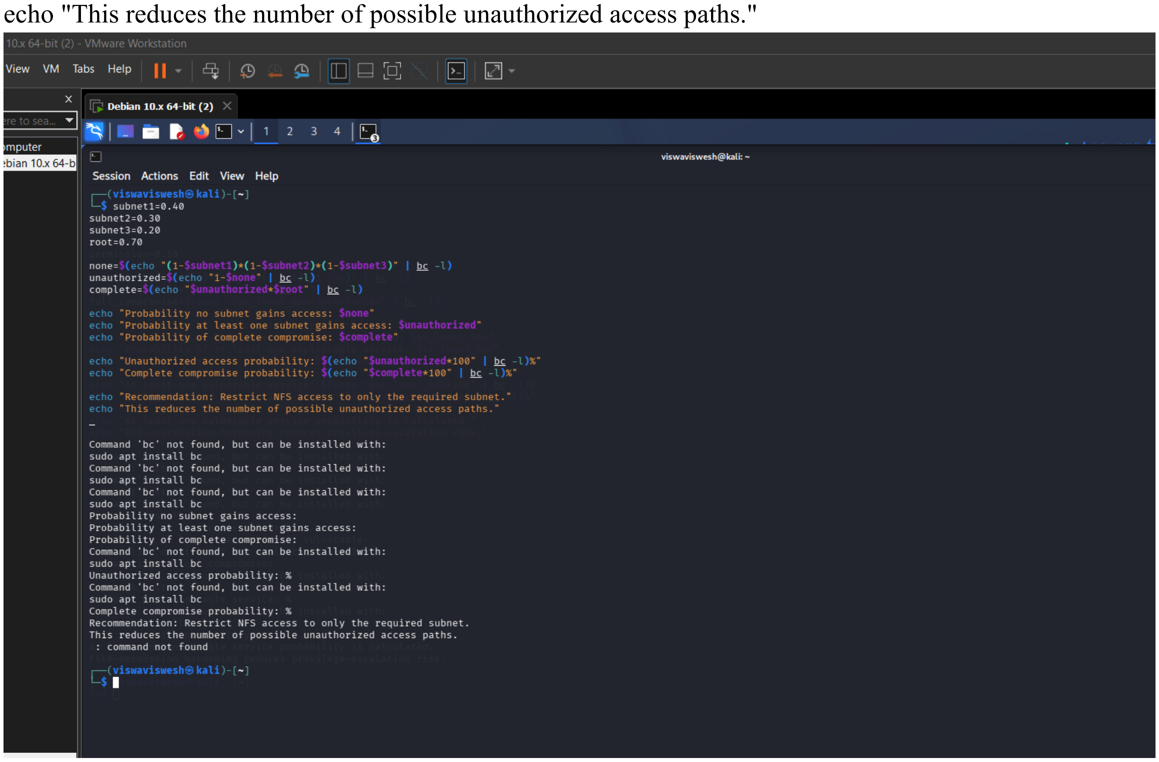
Q3.

```
#!/bin/bash

p1=0.30
p2=0.20
p3=0.25
p4=0.15
permission=0.50

none=$(echo "(1-$p1)*(1-$p2)*(1-$p3)*(1-$p4)" | bc -l)
at_least_one=$(echo "1-$none" | bc -l)
full_compromise=$(echo "$at_least_one*$permission" | bc -l)

echo "Probability no service is vulnerable: $none"
echo "Probability at least one service is vulnerable: $at_least_one"
echo "Probability of at least one exploitable service: $at_least_one"
echo "Probability of full compromise: $full_compromise"

echo "At least one vulnerable service: $(echo "$at_least_one*100" | bc -l)%"
echo "Full compromise probability: $(echo "$full_compromise*100" | bc -l)%"

echo "At least one vulnerable service probability is calculated."
echo "File-permission hardening reduces privilege-escalation risk."
```



Q4. #!/bin/bash

```
subnet1=0.40
subnet2=0.30
subnet3=0.20
root=0.70

none=$(echo "(1-$subnet1)*(1-$subnet2)*(1-$subnet3)" | bc -l)
unauthorized=$(echo "1-$none" | bc -l)
complete=$(echo "$unauthorized*$root" | bc -l)

echo "Probability no subnet gains access: $none"
echo "Probability at least one subnet gains access: $unauthorized"
echo "Probability of complete compromise: $complete"

echo "Unauthorized access probability: $(echo "$unauthorized*100" | bc -l)%"
echo "Complete compromise probability: $(echo "$complete*100" | bc -l)%"

echo "Recommendation: Restrict NFS access to only the required subnet."
echo "This reduces the number of possible unauthorized access paths."
```



Q5. #!/bin/bash

```
records=50000
sensitive_percent=0.30
rate=500
block_time=40

extracted=$(echo "$rate * $block_time" | bc)
percentage=$(echo "$extracted / $records * 100" | bc -l)
sensitive=$(echo "$extracted * $sensitive_percent" | bc)

echo "Total database records: $records"
echo "Extraction rate: $rate records/second"
echo "Attack duration: $block_time seconds"
echo "Records extracted: $extracted"
echo "Percentage of database compromised: $percentage%"
echo "Estimated sensitive records exposed: $sensitive"

if (( $(echo "$percentage >= 30" | bc -l) )); then
   echo "Severity: HIGH"
else
   echo "Severity: MODERATE"
fi

echo "Detection helps limit the attack."
echo "SQL injection should be prevented through secure input handling."
```